

One-Page Reflection: Bayesian vs. MLE Decision Making

Concrete Example: Small-Sample Segment Targeting

Imagine a scenario where a new niche marketing segment (e.g., 'Young Professionals in Rural Areas') shows a 60% churn rate based on only 5 customers ($k=3, n=5$).

Framework	Core Mechanism	Business Action / Decision
Maximum Likelihood Estimation (MLE)	Relies purely on the point estimate from observed data. The churn rate is calculated exactly as 0.60 (60%).	Target them now: Flags the segment as 'High Risk' and authorizes expensive, aggressive discount campaigns to prevent further churn based on a brittle number.
Bayesian Estimation	Incorporate a global prior (e.g., average churn is ~25%). The MAP estimate pulls toward the prior due to the small sample size, explicitly revealing a wide 94% Credible Interval (HDI) spanning 15% to 85%.	Wait for more data: Acknowledges that the 60% rate is likely a high-variance fluke of small numbers, saving the budget until more data clarifies the actual risk.

The Mechanism of Change

The Bayesian framework changes the decision from *"Target them now"* to *"Wait for more data."* The 'mechanism' is the explicit representation of uncertainty. While the MLE provides a single, brittle number that suggests a crisis, the Bayesian posterior reveals that our knowledge of this segment is currently too thin to justify a costly intervention. By acknowledging that the 60% observed rate is likely a high-variance fluke of small numbers, we save the company budget that would have been wasted on an unverified assumption.